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Build a Virtual Private Cloud (VPC)



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VPC > Your VPCs > Create VPC

Create VPC Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create Info
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

NextWork VPC

IPv4 CIDR block Info
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block Info
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy Info
Default



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC (Virtual Private Cloud) is a service that allows you to create a logically isolated section of the AWS cloud, providing you with a virtual network to launch your AWS resources. It gives you control over your virtual networking environment, including IP address ranges, subnets, route tables, and security groups. This means you can tailor your network to your specific needs, enhancing security, flexibility, and control over your AWS infrastructure.

How I used Amazon VPC in this project

I set up a VPC with public subnet. I attached an internet gateway to this VPC for internet connectivity.

One thing I didn't expect in this project was...

Every thing went fine.

This project took me...

30 minuts



Virtual Private Clouds (VPCs)

VPCs are Virtual Private Cloud, a container for AWS resources like EC2 instances etc.

There was already a default VPC in my account ever since my AWS account was created. This is because AWS wants us to have a default container in which we can put our resources like EC2 Instance. Though we can create a custom VPC as well that might suit our needs.

Start your response with 'To set up my VPC, I had to define an IPv4 CIDR block, which is a way to assign a whole block of IP addresses, kind of like creating a zone/area in a city. To understand how big a CIDR block is, look at the number after the slash - the smaller the number, the larger the CIDR block!



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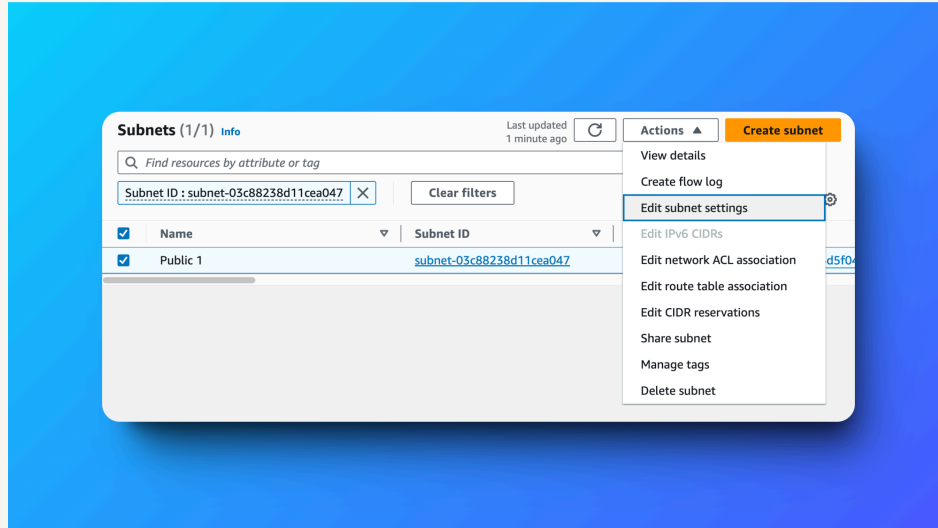


Subnets

Subnets are subsets of VPC spread over AZs in an AWS region. There are already subnets existing in my account, one for every AZ.

Once I created my subnet, I enabled "auto-assign public IPv4 addresses". This setting makes sure that my resource in this subnet gets a public IP so they can reach internet.

The difference between public and private subnets are: A public subnet is connected to the internet ☐ Resources inside a public subnet can communicate with external networks. A private subnet does not have direct internet access ☐ You'd use it for internal resources that don't need to be publicly accessible. For a subnet to be considered public, it has to an internet gateway.





Internet gateways

Internet gateways are key to making applications available on the internet. By attaching an internet gateway, your instances can access the internet and be accessible to external users.

Attaching an internet gateway to a VPC means connecting it to the public internet. If i missed this step, even EC2 instances with Public IPs will not be able to connect to the internet.

VPC > Internet gateways > Create internet gateway

Create internet gateway Info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Create a tag with a key of 'Name' and a value that you specify.

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Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	NextWork IG	Remove

Add new tag

You can add 49 more tags.

Cancel Create internet gateway



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